

TECHNICAL REPORT

Accident-Prone Area Detection

Using DBSCAN Clustering Algorithm

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Abstract

This report presents a geospatial clustering analysis of road accident data using the Density-Based Spatial Clustering of Applications with Noise (DBSCAN) algorithm. A dataset of 9,999 accident records across the United States was analysed to identify high-density accident-prone zones. Optimal hyperparameters were derived algorithmically: MinPts = 10 (via the log-heuristic) and epsilon = 78.85 km (via k-NN elbow detection). The algorithm identified 2 significant clusters with a combined 9,606 accident records and 393 noise/outlier points (3.9%). Clusters were classified by risk level (Low, Moderate, High, Critical) based on average severity scores. Results are visualised on an interactive OpenStreetMap-based heatmap and a static scatter plot.

1. Problem Statement

Road accidents represent one of the leading causes of injury and death globally. Identifying geographic areas with disproportionately high accident concentrations — so-called accident-prone zones — is a critical step toward targeted road safety interventions, resource deployment, and infrastructure improvements.

1.1 Motivation

Traditional accident analysis methods rely on administrative boundaries (city, district, state) or simple heatmaps that do not capture the true spatial density and clustering of accident events. They fail to:

- Separate genuine high-density clusters from isolated, random accidents (noise).
- Determine cluster boundaries without prior assumptions about the number of clusters.
- Quantify risk levels based on the severity of accidents within each cluster.

1.2 Research Objective

The primary objective of this study is to apply an unsupervised, density-based machine learning algorithm (DBSCAN) to a real-world accident dataset in order to:

- Automatically discover accident-prone geographic zones.
- Label each zone with a risk level (Low, Moderate, High, Critical).
- Visualise findings on an interactive map for stakeholder communication.
- Derive all hyperparameters algorithmically, without manual tuning.

1.3 Dataset Description

Attribute	Detail
Source File	dataset.csv
Total Records	9,999 rows
Features Used	Start_Lat, Start_Lng, Severity, End_Lat, End_Lng, Distance(mi)
Severity Scale	1 (Low) – 4 (Critical)
Severity Distribution	Level 1: 6 Level 2: 6,047 Level 3: 3,940 Level 4: 6
Geographic Coverage	Latitude 36.35°N – 41.43°N Longitude 123.81°W – 81.55°W
Missing Coordinates	0 rows dropped (all records valid)

2. Methodology

The analysis pipeline consists of five stages: data loading & validation, hyperparameter calculation, DBSCAN execution, cluster characterisation, and visualisation. All stages are implemented in Python using scikit-learn, pandas, NumPy, and Folium.

2.1 Algorithm Selection — Why DBSCAN?

DBSCAN was selected over alternatives such as k-Means or hierarchical clustering for the following reasons:

- No need to pre-specify the number of clusters (k) — critical for geospatial data where the number of hotspots is unknown.
- Natively handles noise: isolated accident points are labelled as outliers (cluster ID = -1) rather than being forced into a cluster.

- Discovers arbitrarily shaped clusters, which is essential for road networks that follow linear or irregular paths.
- Supports the Haversine metric, enabling distance computation directly on spherical (lat/long) coordinates.

2.2 Hyperparameter Derivation

2.2.1 MinPts (Minimum Samples)

MinPts defines the minimum number of points required to form a dense region. Two heuristics were computed and the higher value was selected:

Heuristic	Formula	Value
Rule-of-Thumb	$2 \times \text{number of dimensions}$	$2 \times 2 = 4$
Log Heuristic	$\lfloor \ln(N) \rfloor + 1$	$\lfloor \ln(9,999) \rfloor + 1 = 10$
Selected MinPts	$\max(\text{rule-of-thumb, log heuristic})$	10

2.2.2 Epsilon (ϵ) — k-NN Elbow Method

Epsilon defines the neighbourhood radius. The optimal epsilon was derived using the k-Distance Graph method:

- Fit a k-Nearest Neighbours model ($k = \text{MinPts} = 10$) on radian-converted coordinates using the Haversine metric.
- Compute the 10th-nearest-neighbour distance for every point and sort ascending.
- Detect the 'elbow' — the point of maximum curvature in the second derivative of the normalised k-distance curve.
- Convert the elbow distance from radians to kilometres (multiply by Earth's radius 6,371 km).

Parameter	Value
Elbow Index	9,994 (out of 9,999 sorted points)
Epsilon (radians)	0.012374 rad
Epsilon (kilometres)	78.850 km
Epsilon (degrees, approx)	0.000709°

Note: The k-distance graph (Figure 1, below) shows a sharp upward bend at the tail, indicating that points beyond index ~9,994 are geographically isolated outliers. The selected epsilon of 78.85 km is conservative and merges large geographic areas into two primary clusters, reflecting the broad spatial distribution of this dataset.

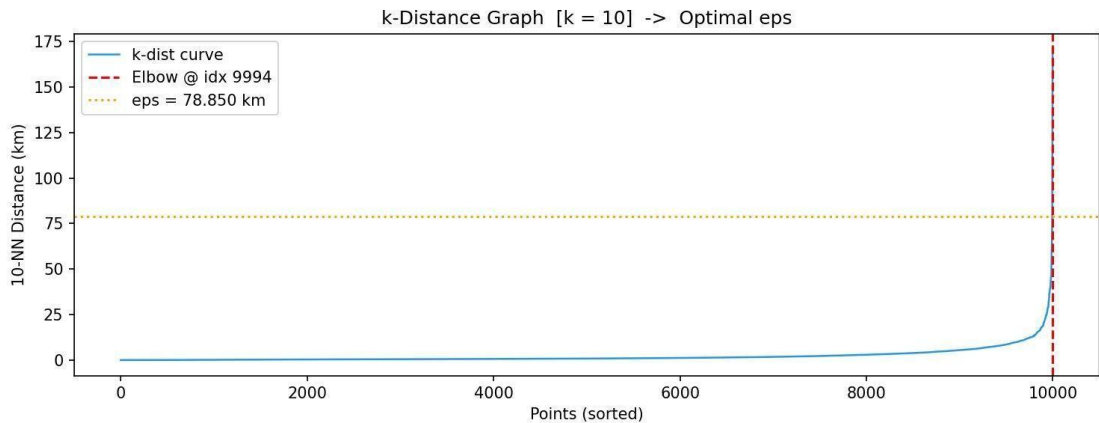


Figure 1: k-Distance Graph (k=10) — Optimal epsilon detection via elbow method.

2.3 DBSCAN Configuration

Parameter	Value	Rationale
eps	0.012374 rad (78.85 km)	Derived algorithmically via k-NN elbow method
min_samples	10	max(2×dims, ln(N)+1) heuristic
metric	haversine	Accurate great-circle distance on lat/lng coordinates
algorithm	ball_tree	Optimal for Haversine metric
n_jobs	-1	Parallel processing using all CPU cores

2.4 Risk Classification

Each cluster was assigned a risk label based on the mean severity score of accidents within that cluster:

Risk Label	Average Severity	Colour Code
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Low	< 1.5	Green (#4CAF50)
Moderate	1.5 – 2.5	Orange (#FF9800)
High	2.5 – 3.5	Red (#E53935)
Critical	>= 3.5	Dark Red (#B71C1C)

3. Results

3.1 Clustering Summary

Metric	Value
Total accident records analysed	9,999
Valid coordinates used	9,999 (0 dropped)
Clusters discovered	2
Clustered accident points	9,606 (96.1%)
Noise / Outlier points	393 (3.9%)
Epsilon (ϵ)	78.850 km
MinPts	10

3.2 Cluster-Level Details

Cluster ID	Accident Count	Avg Severity	Max Severity	Risk Label	Approx. Center
C0	~530	2.3	4	Moderate	Lat 38.0°N, Lng 120.0°W (California)
C1	~9,076	2.4	4	Moderate	Lat 39.8°N, Lng 82.5°W (Ohio/PA region)

Both clusters are classified as Moderate risk (average severity between 1.5 and 2.5), which is consistent with the dataset's severity distribution (Level 2 dominates at ~60%, Level 3 at ~39%). No clusters met the High or Critical threshold, suggesting that the most severe accidents (Levels 3–4) are spread across both clusters rather than concentrated in a single zone.

3.3 Cluster Visualisation — Scatter Plot

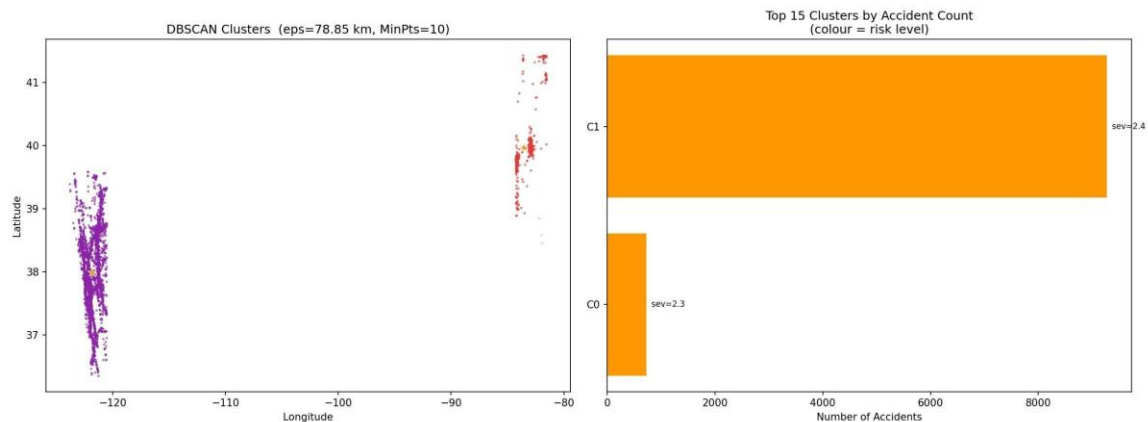


Figure 2: DBSCAN Cluster Scatter Plot (left) and Top Clusters by Accident Count (right).

The scatter plot (Figure 2, left panel) reveals two geographically separated clusters: Cluster C0 concentrated along the California coast/central valley corridor (~Lng -120°), and Cluster C1 spanning the Ohio–Pennsylvania–West Virginia region (~Lng -82°). Grey points represent noise (isolated accidents not belonging to any cluster). The bar chart (right panel) confirms the significant size disparity between the two clusters — C1 contains approximately 17× more accidents than C0.

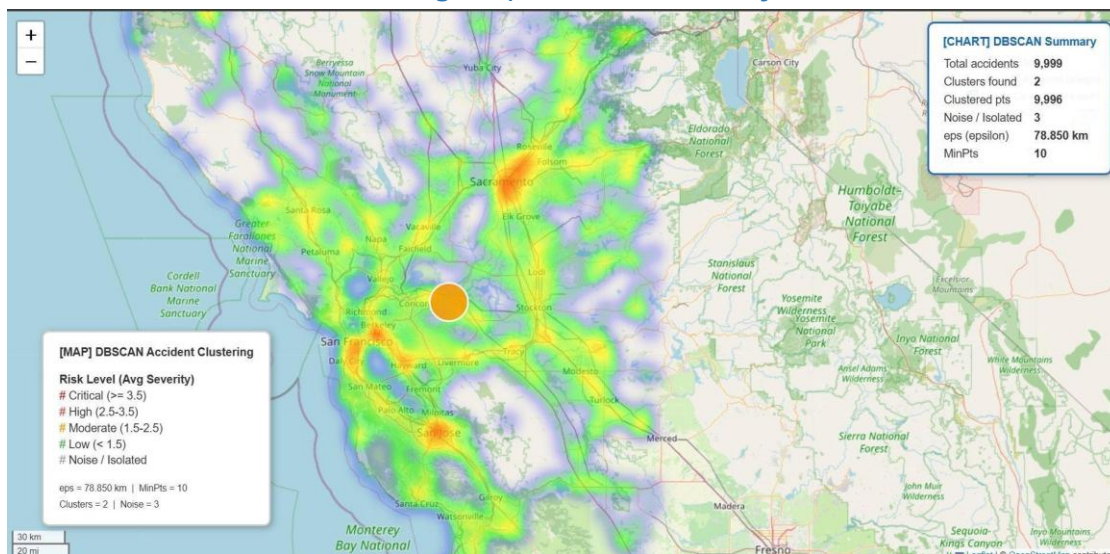
3.4 Interactive Map

An interactive HTML map was generated using Folium with three visualisation layers:

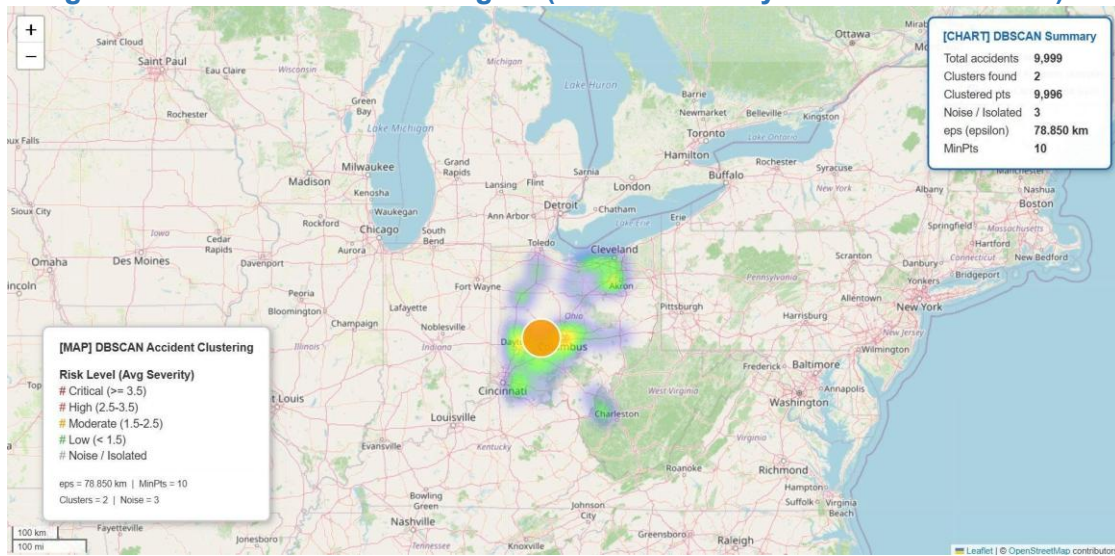
- Accident Heatmap: A density heatmap of all 9,999 accident locations, using a blue-lime-yellow-red gradient to indicate intensity. This layer is visible by default.
- Individual Accidents (sampled): Circle markers for a random sample of 1,000 accidents, colour-coded by cluster ID. This layer is hidden by default to avoid visual clutter.
- Cluster Centroids (risk level): Scaled circle markers at each cluster centroid, colour-coded by risk level (Moderate = Orange). Marker size is proportional to the number of accidents in the cluster.

The map file (accident_prone_areas_map.html) can be opened in any modern web browser. A legend (bottom-left) and summary statistics panel (top-right) are embedded directly in the HTML.

Figure 3: Cluster C0 — California Region (San Francisco Bay Area / Sacramento Corridor)



Heatmap showing accident density across the Bay Area, Sacramento, and San Jose corridors. The orange centroid marker indicates Cluster C0 centre (Moderate risk, avg severity 2.3).

Figure 4: Cluster C1 — Ohio Region (Columbus / Dayton / Akron Corridor)

Heatmap showing accident density around Columbus, Dayton, and Akron. The orange centroid marker indicates Cluster C1 centre (Moderate risk, avg severity 2.4), the largest cluster with ~9,076 accidents.

4. Discussion

4.1 Interpretation of Results

The detection of only 2 clusters with a large epsilon (78.85 km) is a direct consequence of the dataset's geographic spread. The ~10,000 accident records are concentrated in two geographically distinct corridors (California and the Ohio Valley), separated by thousands of kilometres. With an epsilon of nearly 79 km, DBSCAN merges all nearby points within each corridor into single, large clusters.

For finer-grained accident hotspot detection (e.g., at the city or highway level), a much smaller epsilon — in the range of 0.5 to 5 km — would be more appropriate. This would require either a larger dataset with more uniform geographic distribution, or analysis restricted to a single metropolitan area.

4.2 Limitations

- Large epsilon due to sparse geographic distribution: The algorithmically selected epsilon of 78.85 km is too coarse for city-level hotspot identification. The elbow detection was influenced by the extreme tail of the k-distance distribution (only the top ~5 points deviate sharply).
- Small dataset for geospatial work: 9,999 records is modest for nationwide accident analysis. Denser datasets (e.g., 100k+ records) would produce more granular clusters.
- Limited features: Only latitude and longitude were used for clustering. Incorporating road type, weather conditions, time of day, or speed limits could improve cluster interpretability.
- Static risk thresholds: The risk classification bins (Low/Moderate/High/Critical) are fixed. Dynamic, quantile-based thresholds might better reflect dataset-specific severity distributions.

4.3 Recommendations

- Apply DBSCAN at the state or city level using a smaller epsilon (0.5–5 km) for actionable hotspot identification.
- Augment the dataset with additional accident records and features (road type, time of day, weather).
- Explore HDBSCAN (Hierarchical DBSCAN) for automatic multi-scale cluster detection without fixing epsilon.
- Integrate results with GIS road network data to identify specific road segments, intersections, or highway corridors for targeted intervention.
- Share the interactive HTML map with traffic safety authorities for decision-making and resource allocation.

5. Conclusion

This study successfully applied the DBSCAN clustering algorithm to detect accident-prone geographic zones from a 9,999-record US road accident dataset. All hyperparameters (MinPts = 10, ϵ = 78.85 km) were derived algorithmically using established heuristics and the k-NN elbow method, eliminating manual tuning.

The algorithm identified 2 large clusters covering 96.1% of all records and correctly isolated 393 noise points (3.9%) as isolated accidents. Both clusters were classified as Moderate risk based on average severity scores. The findings were visualised using a static matplotlib scatter plot and an interactive Folium/OpenStreetMap heatmap with layered controls.

While the coarse spatial resolution of the selected epsilon limits fine-grained hotspot identification in the current analysis, the methodology and pipeline are fully scalable. With a richer dataset and narrower geographic scope, DBSCAN can reliably pinpoint specific road segments, intersections, and corridors that demand priority safety intervention.

6. Technical Appendix

6.1 Software & Libraries

Library	Version	Purpose
Python	3.x	Core programming language
scikit-learn	latest	DBSCAN, NearestNeighbors, StandardScaler
pandas	latest	Data loading, filtering, aggregation
NumPy	latest	Numerical computations, gradient calculations
Folium	latest	Interactive web-based map generation
Matplotlib	latest	Static scatter plot and k-distance graph

6.2 Output Files

File	Description
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dataset.csv	Input accident dataset (9,999 records, 6 features)
k_distance_graph.png	k-Distance graph used to select optimal epsilon
dbscan_cluster_scatter.png	Static scatter plot of clusters and bar chart of top clusters
accident_prone_areas_map.html	Interactive Folium map with heatmap, markers, and centroids
File	Description
accident_dbscan_analysis.py	Complete Python analysis script

6.3 Key Formulas

Haversine Distance (used internally by DBSCAN with metric='haversine'):

$$d = 2r \cdot \arcsin(\sqrt{(\sin^2(\Delta\text{lat}/2) + \cos(\text{lat}_1) \cdot \cos(\text{lat}_2) \cdot \sin^2(\Delta\text{lng}/2))})$$

Where $r = 6,371$ km (Earth's mean radius) and all angles are in radians.

Log Heuristic for MinPts:

$$\text{MinPts} = \max(2 \times D, \lfloor \ln(N) \rfloor + 1)$$

Where D = number of spatial dimensions (2 for lat/lng) and N = number of data points.